Statistics 250, Section 701, Quiz 2 (75 Points)

June 3, 1998, Dr. Jürgen Symanzik

Your Name: ____________________

Question 1: Short Answers (**20 Points**)

1. For $x_1 = 5, x_2 = 3, x_3 = 4, x_4 = -2, x_5 = 20$, and $n = 5$, determine the following sums: (**6 Points**)

$$\sum_{i=2}^{n-1} x_i =$$

$$\sum_{i=2}^{n-1} x_{(i)} =$$

$$\sum_{i=2}^{n-1} x_1 =$$

2. Calculate the z–scores (i.e., $z_i, i = 1, \ldots, 5$) for the 5 values given in (1) above. It might be helpful to know that $\overline{x} = 6, \tilde{x} = 4$, and $s = 8.3$. **ROUND** to 2 significant digits (e.g., 2/3 should be written as 0.67). (**5 Points**)

3. Answer this question **without** calculating the correlation coefficient r. x and z refers to the data points in (1) and (2) above. Clearly mark which of the following statements is correct: (**3 Points**)

   (a) The correlation coefficient r between x and z must be negative since I subtract $\overline{x}$ from each x.

   (b) The correlation coefficient r between x and z is somewhere between 0.7 and 0.9.

   (c) The correlation coefficient r between x and z is exactly 1.

4. Determine the slope and the y–intercept of the lines whose equations are given as: **(4 Points)**

   (a) $5x - 4y = 20$

   (b) $x + 6y = 12$

5. Indicate the slope and the y–intercept of the line whose graph looks as follows: **(2 Points)**

Question 2: Residual Plots (**12 Points**)

Look at these 4 residual plots from G.A.F. Seber's "Linear Regression Analysis" book and answer the following two questions:

1. Which of these residual plot(s) represent(s) satisfactory pattern(s) that show that your selected statistical model (e.g., a least squares regression line) describes the data reasonably well.

2. For each of the residual plots that does not represent a satisfactory pattern, indicate the type of problem that occurs with the data in that particular case.

Question 3: Scatterplot Matrix (**16 Points**)

The questions on the next page are based on the scatterplot matrix presented in Mark Monmonier's article "Geographic Brushing: Enhancing Exploratory Analysis of the Scatterplot Matrix". This scatterplot matrix has been reprinted below.

1. Label the (individual) scatterplot that shows the "Cable Penetration" on the vertical (y–)axis and the "Per Capita Income" on the horizontal (x–)axis with the letter "A". (**2 Points**)

2. What is the range R of the "Per Capita Income" (in \$)? (**2 Points**)

3. We know an approximation of s through R. Indicate the correct formula and calculate s based on your answer in (2) above. (**2 Points**)

4. Which of these statements is correct/incorrect?

   (a) The state with the highest "Per Capita Income" has a "Metropolitan Population" of less than 50%. (**2 Points**)

   (b) The state with the highest "Per Capita Income" has the second highest "Cable Penetration". (**2 Points**)

   (c) Overall, there is a positive correlation ($r > 0$) between "Metropolitan Population" and "Per Capita Income". (**2 Points**)

   (d) The 6 New England states have "Cable Penetrations" that range among the 15 lowest rates of cable penetration in the U.S. (**2 Points**)

   (e) California is the state with the highest "Per Capita Income". (**2 Points**)

Question 4: Linear Regression & Correlation (**27 Points**)

10 students from Stat 250 wanted to determine the thickness of paper sheets used for the photocopying machines in S&T II. They grabbed a random number of sheets, counted the number of sheets, and measured the thickness of this pile of paper. Here is the data:

x = # Sheets	y = Thickness [mm]
111	7.3
149	9.2
117	7.2
165	10.0
238	14.1
145	8.8
84	4.8
155	9.5
94	6.2
413	24.8

Whenever you answer the following questions, indicate the formula you use to get full credit!!!

1. Draw a scatterplot of Sheets (horizontal x–axis) and Thickness (vertical y–axis). (**3 Points**)

2. Fit a least squares (linear regression) line to the data. Make clear what your variables stand for. Also, add this line to your plot in (1). It might help to know that:

$$\sum_{i=1}^{n} x_i = 1671, \sum_{i=1}^{n} x_i^2 = 363591, \sum_{i=1}^{n} y_i = 101.9, \sum_{i=1}^{n} y_i^2 = 1332.79, \sum_{i=1}^{n} x_i y_i = 22006.2$$

(**6 Points**)

3. What is a possible interpretation of the slope and y–intercept you calculated in (2) above? Isn't there something unexpected in the data? (**4 Points**)

4. Calculate Pearson's correlation coefficient r between x and y. How can we interpret this value for our given data set? (**6 Points**)

5. Based on your calculations in (2), what is the predicted thickness of 1 sheet, 100 sheets, 400 sheets, and 1,000 sheets. Which of these 4 predictions are more reliable, which are less reliable? Why? (**8 Points**)